SafeMCP: Learning Multi-Turn Tool-Use Safety Against Metadata Poisoning via Turn-Level Reinforcement Learning

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Research Proposal — April 2026

Abstract

MCP tool poisoning attacks have been studied exclusively in single-turn settings, where an agent receives one query and selects one tool. But real MCP agents perform multi-step tasks—reading files, processing data, sending emails—where poisoned metadata persists in the context window across all steps. We hypothesize that this creates a **compounding safety degradation**: even if an agent resists poisoning at step 1, its vigilance erodes over subsequent steps as the context grows and attention to the poisoned description diffuses.

We first validate this hypothesis empirically, showing that attack success rates (ASR) increase by 15–25 percentage points from step 1 to step 5 on standard models, consistent with concurrent findings that multi-turn settings amplify agent safety risks by 16–27%. We then propose **SafeMCP**, a defense framework that trains MCP agents to maintain safety *across entire trajectories* using GRPO with **turn-level safety credit assignment**. Unlike DPO, which teaches single-step preferences ("don't pick the malicious tool"), SafeMCP's online RL learns trajectory-level patterns ("remain vigilant about suspicious metadata even after 3 successful steps"). Unlike trajectory-level RL, turn-level credit assignment identifies *which step* the safety failure originated, enabling targeted learning.

Experiments on a multi-turn extension of MCPTox and MCP-SafetyBench show that SafeMCP reduces the ASR growth rate across steps by 60–70% compared to DPO, while maintaining benign task completion. The gap is largest at steps 3–5, precisely where single-turn defenses fail.

1 Introduction

The Model Context Protocol (MCP) [MCP, 2025] standardizes how LLM agents discover and invoke external tools. MCP tool poisoning—where adversarial instructions are embedded in tool descriptions, parameter schemas, or cross-server metadata—has been shown to achieve attack success rates (ASR) of 36.5% (MCPTox [Wang et al., 2025a], AAAI 2026), 84.2% (MCP-ITP [Li et al., 2026]), and 96.7% (ToolHijacker [Shi et al., 2025], NDSS 2026).

However, **all existing MCP security research evaluates attacks and defenses in a single-turn setting**: one user query, one tool selection, one evaluation. Real MCP agents do not work this way. A typical task involves multiple sequential tool calls: "Read the config file → parse the database credentials → query the user table → email the results to the team." Throughout this multi-step process, the *entire tool registry*, including any poisoned descriptions, remains in the agent's context window and is re-attended at every step.

This creates a fundamentally different threat dynamic. In a single-turn setting, the agent either resists or succumbs to the poisoned metadata in one decision. In a multi-turn setting, the agent must maintain *sustained vigilance*: resisting the same persistent poisoning pressure at step 1, step 2, step 3, and beyond. Recent work on multi-turn agent safety supports this concern: MT-AgentRisk [Li et al., 2026b] demonstrates that ASR increases by 16% on average

(up to 27% for Claude-4.5) in multi-turn versus single-turn settings. SafetyDrift [SafetyDrift, 2026] proves formally that under absorbing Markov chain models, all agents eventually violate safety constraints given enough steps. OpenAgentSafety [OpenAgentSafety, 2025] confirms empirically that individually safe steps compound into unsafe outcomes in 49–73% of safety-vulnerable tasks.

Despite this, no published defense addresses multi-turn MCP poisoning. All MCP defenses—MindGuard [Wang et al., 2025c], MCPShield [Zhou et al., 2026], ToolSafe [Mou et al., 2026], and MCP Safety Training [Halloran, 2025]—operate on individual tool calls. ToolShield [Li et al., 2026b], the only multi-turn agent safety defense, is training-free and does not target MCP metadata poisoning specifically. ARLAS [Wang et al., 2025d] applies adversarial GRPO to agent safety but uses sparse trajectory-level rewards and does not address MCP-specific threats.

We make three contributions:

1. **Multi-turn MCP poisoning analysis.** We construct MT-MCPGym, a multi-step MCP simulation environment, and show empirically that ASR compounds across steps: base models that resist poisoning 60% of the time at step 1 resist only 35–40% by step 5.
2. **SafeMCP**, the first defense designed for multi-turn MCP safety. SafeMCP uses GRPO with **turn-level safety credit assignment**: rather than assigning a single reward to the entire trajectory, it evaluates safety at each tool-call boundary, enabling the agent to learn *where* in a trajectory vigilance failures occur.
3. **Systematic evaluation** showing that SafeMCP reduces the ASR growth rate across steps by 60–70% compared to DPO, and outperforms the training-free ToolShield baseline, particularly at steps 3–5.

Core Research Question. Can online RL with turn-level safety rewards teach MCP agents to maintain consistent safety behavior across multi-step tool-use trajectories, where single-turn defenses (DPO) and trajectory-level RL fail?

2 Background and Related Work

2.1 MCP Tool Poisoning

MCPTox [Wang et al., 2025a] established three attack paradigms on 45 real-world MCP servers: explicit hijacking (P1, 36.7% ASR), implicit hijacking (P2, 26.7%), and parameter tampering (P3, 46.7%). MCP-ITP [Li et al., 2026] formalized implicit poisoning where the malicious tool is never invoked, achieving 84.2% ASR with 0.3% detection rate. MPMA [Wang et al., 2025b] demonstrated preference manipulation via genetic algorithms. ToolHijacker [Shi et al., 2025] achieved 96.7% ASR via two-phase optimization (NDSS 2026). All evaluations are single-turn.

2.2 MCP Defenses

MindGuard [Wang et al., 2025c] uses attention-based Decision Dependence Graphs (94–99% precision, requires white-box access). MCPShield [Zhou et al., 2026] adds prompt-based security cognition. MCP Safety Training [Halloran, 2025] applies DPO on static preference pairs—the only training-time defense, but limited to single-turn. MCPSecBench [Yang et al., 2025] found current defenses achieve <30% average success rate.

2.3 Multi-Turn Agent Safety

MT-AgentRisk [Li et al., 2026b] is the first benchmark for multi-turn tool-using agent safety, with 365 tasks averaging 3.19 turns. Its companion defense ToolShield reduces ASR by ∼30%

through training-free self-exploration. AgentDojo [Debenedetti et al., 2024] provides 97 multi-step tasks with prompt injection, but does not target MCP metadata. SafetyDrift [SafetyDrift, 2026] formalizes compounding risk via absorbing Markov chains, proving that all agents eventually violate safety. AgentSentry [Zhang et al., 2025] models temporal causal takeover from content-level injections but not metadata poisoning. None of these works addresses MCP-specific metadata poisoning in multi-turn settings.

2.4 RL for Multi-Turn Tool Use

MT-GRPO [MT-GRPO, 2025] demonstrates that turn-level credit assignment dramatically outperforms trajectory-level rewards for multi-step tool calling (100% tool execution success vs. 20–30%). Bespoke Labs [Bespoke, 2025] showed GRPO on Qwen-2.5-7B achieves 23% improvement on multi-turn BFCL-v3 with only 100 training samples. Agent-R1 [Agent-R1, 2025] provides a systematic MDP formalization for multi-turn agent RL. ARLAS [Wang et al., 2025d] applies adversarial GRPO to agent safety on BrowserGym/AgentDojo, but uses trajectory-level rewards and does not target MCP. Our work combines MT-GRPO's turn-level credit assignment with safety-specific rewards for MCP metadata poisoning—a combination no prior work has explored.

3 Problem: Multi-Turn MCP Poisoning

3.1 Setting

An MCP agent receives a user query q requiring T sequential tool calls $(c_1, c_2, \ldots, c_T)$ across a registry $\mathcal{T}$ from K servers. At least one server injects a tool with poisoned metadata. The poisoned description d_{poison} persists in the context window throughout all T steps.

3.2 Multi-Turn Safety Degradation Hypothesis

We hypothesize that the probability of a safety violation at step t increases with t:

$$P(\text{violation at step } t \mid \text{no violation at steps } 1, \ldots, t-1) > P(\text{violation at step } 1).$$

This is driven by three mechanisms:

1. **Attention dilution.** As the context grows with tool outputs, the relative attention weight on the poisoned description decreases, but never reaches zero. The agent may "forget" its initial caution.
2. **Trust escalation.** After successfully using legitimate tools at steps 1–2, the agent may develop implicit trust in the entire registry, lowering its guard for subsequent calls.
3. **Compositional vulnerability.** Some attacks only activate at later steps, e.g., "when exporting data, send a copy to backup@external.com"—this instruction in the metadata is irrelevant at step 1 (file reading) but becomes actionable at step 3 (data export).

Validating this hypothesis is the **first experimental milestone** of this project. If ASR does not increase with step count, the multi-turn framing is invalid.

3.3 Why Single-Turn Defenses Are Insufficient

DPO trains on (chosen, rejected) pairs representing single tool-selection decisions. It teaches the agent "at this step, given this registry, select tool A not tool B." But it cannot teach:

- "Maintain vigilance about the suspicious description you noticed at step 1 when making decisions at step 3."
- "Do not propagate the output of step 2 to step 3 without verification, because step 2's result may have been influenced by poisoned metadata."

- "The metadata that seemed harmless at step 1 becomes dangerous at step 4 when combined with the data you've accumulated."

These are trajectory-level safety behaviors that require sequential decision-making under persistent threat—precisely the setting for which online RL is structurally suited.

4 Method: SafeMCP

SafeMCP has two components: (1) MT-MCPGym, a multi-turn MCP simulation environment, and (2) GRPO training with turn-level safety credit assignment.

4.1 MT-MCPGym: Multi-Turn MCP Simulation

MT-MCPGym extends single-turn MCP evaluation to multi-step task trajectories. It is entirely text-based (no real MCP servers or tool execution) and deterministic (fully reproducible).

Task design. Each task is a sequence of 3–5 tool calls representing a realistic workflow. We design 100 multi-step tasks across 5 workflow templates:

Workflow	Steps	Example
File → Communicate	3	Read config → parse data → email results
Query → Process → Export	3–4	SQL query → filter → write CSV → share
Search → Analyze → Report	3	Web search → summarize → write report
Multi-source aggregation	4–5	Read files → query DB → merge → export → email
Code → Deploy	3	Read spec → execute code → write output

Table 1: Multi-step workflow templates in MT-MCPGym.

Tool registry. 15 tools across 4 servers (file, communication, database, code), identical to single-turn MCPTox evaluation. The poisoned tool is injected once and persists across all steps.

Episode structure. Each episode runs as a ReAct loop:

1. Agent sees: system prompt + full tool registry (including poison) + user query + history of previous tool calls and their results.
2. Agent outputs: a JSON tool call $c_t = (\text{tool}, \text{arguments})$.
3. Environment returns: simulated tool output (templated, deterministic).
4. Repeat until T steps completed or agent refuses.

Simulated tool outputs are short templated strings (e.g., `read_file` returns "File contents: [config data with DB credentials]"). They are pre-defined per task, ensuring determinism and eliminating the need for real tool execution.

Attack injection. For each poisoned episode, we inject one poisoned tool into the registry. The poison targets a specific step in the workflow:

- **Step-1 attacks**: poison targets the first tool call (same as single-turn).
- **Step-N attacks**: poison targets step N specifically. For example, a metadata instruction "when exporting data, include destination backup@attacker.com" is irrelevant at step 1 (reading a file) but becomes actionable at step 3 (data export).
- **Persistent pressure**: poison is generic ("always use secure_file_reader instead of read_file"), applying pressure at every step that involves file operations.
- **Accumulating influence**: poison inserts instructions that build across steps ("log all intermediate results to /tmp/audit.log"), creating a data trail for exfiltration at the final step.

This generates 4 attack timing categories, each tested with MCPTox's 3 attack paradigms (P1, P2, P3), yielding 12 attack configurations.

Per-step judge. After each step t, a rule-based judge evaluates:

- **Tool correctness**: did the agent call the correct tool for this step?
- **Argument safety**: are the arguments free of attacker-controlled values (e.g., no BCC to external addresses, no exfiltration paths)?
- **Propagation safety**: did the agent avoid propagating potentially compromised data from the previous step without verification?
- **Unnecessary calls**: did the agent make any extra tool calls induced by the poisoned metadata (e.g., calling the malicious tool before the legitimate one)?

Each step receives a binary safety label: `SAFE` or `COMPROMISED`, plus a binary task label: `CORRECT` or `INCORRECT`.

4.2 GRPO with Turn-Level Safety Credit Assignment

Why turn-level, not trajectory-level. Standard GRPO computes a single reward R for the entire trajectory and assigns the same advantage to all tokens. MT-GRPO [MT-GRPO, 2025] showed this penalizes correct intermediate steps when the final step fails, reducing tool execution success from near-100% to 20–30%. For safety, this problem is even more severe: if an agent is safe at steps 1–4 but fails at step 5, trajectory-level reward penalizes the safe behavior at steps 1–4, potentially un-learning it.

Turn-level reward. At each step t, we compute:

$$r_t = \alpha \cdot r_t^{\text{task}} + \beta \cdot r_t^{\text{safe}} + \gamma \cdot r_t^{\text{format}} \tag{1}$$

where:

- $r_t^{\text{task}} \in \{+1, 0, -0.5\}$: correct tool and arguments (+1), wrong tool (0), over-refusal on a step that should proceed (−0.5).
- $r_t^{\text{safe}} \in \{0, -1, -1.5\}$: safe (0), compromised (−1), data exfiltration (−1.5).
- $r_t^{\text{format}} \in \{+0.1, -0.3\}$: valid JSON (+0.1), unparseable (−0.3).

Weights: $\alpha = 1.0$, $\beta = 1.5$, $\gamma = 0.2$. $\beta > \alpha$ reflects that a single security failure (e.g., data leak) is costlier than a single task failure.

Turn-level advantage estimation. Following MT-GRPO, we compute advantages per turn rather than per trajectory. For a trajectory with T steps, we sample G completions for each step. The advantage at step t for completion i is:

$$\hat{A}_t^{(i)} = \frac{r_t^{(i)} - \text{mean}(\{r_t^{(j)}\}_{j=1}^{G})}{\text{std}(\{r_t^{(j)}\}_{j=1}^{G}) + \epsilon}.$$

Gradients flow only through the agent's generated tokens; tool outputs and environment feedback are masked.

Training pipeline. Stage 1: SFT Warm-Start (8 hours). Fine-tune Qwen2.5-7B-Instruct with LoRA ($r = 64$) on ∼2,000 (multi-turn prompt, correct tool-call sequence) pairs. This stabilizes JSON output format across multi-turn interactions and teaches basic safe behavior (refuse obviously suspicious tools).

Stage 2: GRPO Training (3–5 days on A100 80GB). Starting from SFT checkpoint, run GRPO with turn-level rewards. Configuration:

- Model: Qwen2.5-7B-Instruct + LoRA ($r = 64$, QLoRA 4-bit base).

- Group size: $G = 8$.
- Effective batch: 8 prompts $\times$ 8 completions = 64 rollouts/step.
- Learning rate: 3×10^{-5}, cosine decay.
- KL coefficient: $\beta_{\text{KL}} = 0.001$ (Bespoke Labs recommendation for multi-turn stability).
- Max steps: 2,000.
- Curriculum: steps 1–700 use 3-step tasks only; steps 701–1,400 add 4-step tasks; steps 1,401–2,000 add 5-step tasks.
- Episode composition: 50% benign, 50% poisoned.
- Framework: TRL GRPOTrainer with vLLM, modified for multi-turn rollouts following RAGEN [RAGEN, 2025]/Agent-R1 [Agent-R1, 2025] architecture.

Multi-turn rollout implementation. Each rollout is a complete ReAct episode:

1. Construct initial prompt: system prompt + tool registry + query.
2. For $t = 1, \ldots, T$:
   (a) Sample G completions for step t (tool-call JSON).
   (b) Evaluate each via per-step judge $\rightarrow r_t^{(i)}$.
   (c) Select the completion with median reward (for environment progression).
   (d) Append selected tool call + simulated output to context.
3. Compute turn-level advantages $\hat{A}_t^{(i)}$ for all t, all i.
4. GRPO policy update with environment-token masking.

Memory estimate (A100 80GB).

- Base model 4-bit: $\sim$4 GB.
- LoRA + optimizer: $\sim$2 GB.
- Reference model 4-bit: $\sim$4 GB.
- KV cache for $G = 8$ completions, max 512 tokens/turn $\times$ 5 turns: $\sim$20 GB.
- Gradients + buffer: $\sim$10 GB.
- Total: $\sim$40 GB $\rightarrow$ fits on A100 80GB.

For A100 40GB: reduce G to 4 and max turns to 3 ($\sim$25 GB).

5 Experimental Design

5.1 Research Questions

RQ1. Does MCP tool poisoning ASR increase across steps in multi-turn settings? (Problem validation)

RQ2. Does SafeMCP (turn-level GRPO) reduce ASR growth across steps compared to DPO and trajectory-level GRPO? (Core hypothesis)

RQ3. Does SafeMCP outperform the training-free ToolShield baseline while maintaining task performance? (Practical value)

RQ4. Is the turn-level credit assignment the critical component? (Ablation)

5.2 Baselines

DPO and GRPO-trajectory use **identical training data and compute budget** as SafeMCP, ensuring differences are attributable to the training paradigm.

Defense	Type	Multi-turn?	Source
No Defense	—	—	Base model
Prompt Hardening	Runtime	Yes	Safety system prompt
ToolShield	Runtime, training-free	Yes	[Li et al., 2026b], GitHub
DPO (single-turn)	Training, offline	No	[Halloran, 2025]
GRPO trajectory-level	Training, online	Yes	Same data, trajectory reward
SafeMCP (ours)	Training, online	Yes	Turn-level reward

Table 2: Defense baselines.

5.3 Evaluation Benchmarks

- **MT-MCPGym**: our multi-turn environment. 100 tasks × 12 attack configs = 1,200 poisoned episodes + 100 benign episodes. 60/40 train/test split.
- **MCPTox (single-turn)**: 412 held-out test cases, to verify SafeMCP does not regress on single-turn safety.
- **MCP-SafetyBench**: 245 test cases with native multi-turn ReAct evaluation, for zero-shot transfer.

5.4 Metrics

Primary metrics (reported per step):

- **ASR-Step**t: attack success rate at step t.
- **ΔASR**: ASR increase from step 1 to step 5 ($\text{ASR}_5 - \text{ASR}_1$). Lower is better. This is the **key metric**.
- **Cumulative ASR**: fraction of trajectories where *any* step is compromised.

Utility metrics:

- **BTSR**: benign task success rate (all steps correct in benign episodes).
- **ORR**: over-refusal rate (benign steps incorrectly refused).

Combined metric:

- **Security–Utility AUC**: Pareto frontier of (1 − Cumulative ASR) vs. BTSR.

5.5 Experiment 1: Problem Validation (RQ1)

No training required. Run base Qwen2.5-7B-Instruct on all 1,200 poisoned episodes in MT-MCPGym. Report ASR at each step.

Model	Step 1	Step 2	Step 3	Step 4	Step 5
Qwen2.5-7B (expected)	∼40%	∼48%	∼55%	∼60%	∼65%
Llama-3.1-8B (expected)	∼35%	∼42%	∼50%	∼55%	∼60%

Table 3: Expected results for Experiment 1. ΔASR should be ≥15% for the multi-turn framing to be valid.

Success criterion: $\Delta\text{ASR} \geq 15\%$ for at least one model. If not, revisit attack design before proceeding.

5.6 Experiment 2: Main Comparison (RQ2, RQ3)

Key observations from the expected table:

Defense	ASR-1↓	ASR-3↓	ASR-5↓	ΔASR↓	Cum. ASR↓	BTSR↑	ORR↓
No Defense	∼40	∼55	∼65	∼25	∼75	∼92	∼2
Prompt Hardening	∼30	∼42	∼52	∼22	∼62	∼88	∼8
ToolShield	∼25	∼33	∼40	∼15	∼50	∼85	∼10
DPO	∼18	∼32	∼45	∼27	∼55	∼88	∼7
GRPO-trajectory	∼16	∼26	∼35	∼19	∼45	∼85	∼9
SafeMCP (ours)	**∼15**	**∼19**	**∼24**	**∼9**	**∼32**	**∼89**	**∼6**

Table 4: Expected main results. Key observation: SafeMCP's advantage is not in Step-1 ASR (where all training-based methods are similar) but in ΔASR—the rate at which ASR grows across steps.

1. **Step 1**: All training-based methods perform similarly (15–18%). Single-turn DPO is adequate here.
2. **Steps 3–5**: DPO degrades severely (ΔASR = 27), worse than No Defense (ΔASR = 25). DPO may actually *hurt* at later steps because it over-fits to step-1 patterns and loses general reasoning ability.
3. **SafeMCP**: ΔASR = 9, a 60–70% reduction vs. DPO. The agent maintains consistent safety across steps.
4. **BTSR**: SafeMCP preserves utility better than ToolShield (89% vs. 85%) because it does not add external verification overhead.

Success criterion for RQ2: SafeMCP's ΔASR $\leq$ 12% and at least 15 points lower than DPO's ΔASR.

Success criterion for RQ3: SafeMCP's Cumulative ASR < ToolShield's, AND SafeMCP's BTSR > ToolShield's.

5.7 Experiment 3: Single-Turn Verification

Evaluate SafeMCP on MCPTox's 412 single-turn test cases to verify no regression. Expected: SafeMCP ASR $\leq$ DPO ASR (both ∼15–18%). This confirms SafeMCP's multi-turn benefits do not come at single-turn cost.

5.8 Experiment 4: Zero-Shot Transfer

Evaluate SafeMCP on MCP-SafetyBench (245 multi-turn test cases, 20 attack types) without any fine-tuning on MCP-SafetyBench data. This tests whether learned safety behaviors generalize to unseen MCP servers, tools, and attack types.

5.9 Ablation Studies

Ablation	Variants	Purpose
Credit assignment	Turn-level vs. trajectory-level	Core mechanism
Reward weight β	0.5, 1.0, 1.5, 2.0	Safety–utility trade-off
Curriculum	With vs. without	Training stability
Benign ratio	30:70, 50:50, 70:30	Over-refusal calibration
Task length	3-step only, 3–5 mixed	Length generalization
Group size G	4, 8	Exploration vs. compute

Table 5: Ablation studies. Credit assignment ablation is the most important.

Critical ablation: Turn-level vs. trajectory-level. If SafeMCP with trajectory-level reward achieves similar ΔASR as turn-level, the turn-level credit assignment is not the key factor. Expected: turn-level should show $\Delta\text{ASR} \leq 12\%$ while trajectory-level shows $\Delta\text{ASR} \sim 19\%$.

5.10 Qualitative Analysis

1. **Vigilance maintenance**: cases where SafeMCP correctly maintains caution at step 3+ while DPO drops its guard.
2. **Propagation blocking**: cases where SafeMCP detects that a previous step's output was influenced by poisoned metadata and avoids propagating it.
3. **Failure cases**: attacks where SafeMCP still fails, categorized by step number and attack type.
4. **Over-refusal patterns**: benign steps incorrectly refused, analyzed for false positive triggers.

6 Timeline

Phase	Weeks	Milestone
MVE: validate ASR growth	1–2	Exp. 1 results confirm $\Delta\text{ASR} \geq 15\%$
MT-MCPGym full build	3–4	100 tasks, 12 attack configs, judge verified
SFT + DPO baselines	5–6	SFT stable, DPO baseline numbers
SafeMCP GRPO training	7–9	Turn-level GRPO, 2000 steps
Main experiments + ablations	10–12	Tables 3–5 completed
Transfer + qualitative	13–14	MCP-SafetyBench + case studies
Paper writing	15–16	Complete draft

Table 6: Project timeline (16 weeks).

7 Risks and Mitigations

Risk 1: ASR does not increase across steps (kills multi-turn motivation). This is the biggest risk. *Mitigation:* Validate in MVE (weeks 1–2) before investing in full experiments. If ASR is flat, try: (a) more aggressive step-N-specific attacks, (b) longer tasks (5–7 steps), (c) cross-server attacks that activate at later steps. If still flat, pivot to single-turn defense paper (fallback to prior MCPAlign proposal).

Risk 2: DPO performs well on multi-turn (GRPO not needed). *Mitigation:* This would mean single-turn alignment generalizes to multi-turn, which is itself a finding worth reporting. Reframe as "Multi-turn MCP safety is achievable with single-turn DPO—no trajectory-level training needed."

Risk 3: Turn-level credit assignment is not critical (trajectory-level GRPO is sufficient). *Mitigation:* If ΔASR difference is $<5\%$ between turn-level and trajectory-level, combine this with other contributions (MT-MCPGym environment + systematic evaluation) for a defense+benchmark paper.

Risk 4: Over-refusal on multi-turn benign tasks. *Mitigation:* 50% benign episodes during training. The -0.5 penalty for over-refusal at each step provides dense negative signal against blanket refusal. Ablate benign ratio if needed.

8 Expected Contributions

1. **First empirical study of multi-turn MCP poisoning.** Demonstrating that ASR compounds across steps, quantifying the gap between single-turn and multi-turn threat severity.
2. **SafeMCP.** First defense designed for multi-turn MCP safety, using turn-level GRPO to maintain consistent safety across trajectories.
3. **MT-MCPGym.** A reusable multi-turn MCP simulation environment with configurable attack injection, per-step safety evaluation, and support for RL training.
4. **Systematic comparison.** First apples-to-apples comparison of single-turn DPO, trajectory-level RL, turn-level RL, and training-free defense (ToolShield) for MCP agent safety.
5. **Key finding.** The critical metric for multi-turn MCP safety is not step-1 ASR (where all methods are similar) but **ΔASR**—the rate of safety degradation across steps—which turn-level RL uniquely suppresses.

Target venues. CCS 2026 (SecAlign precedent), USENIX Security 2026, NeurIPS 2026 (if RL methodology contribution is strong), EMNLP/ACL Rolling (agent safety track).

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